

Mathematical Foundations of Reinforcement Learning and Online Decision Making

Lecture 7

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Overview

- 1 TD-Learning with Linear Function Approximation
- 2 Policy Gradient

Roadmap

- 1 Policy evaluation
- 2 The TD(0) algorithm
- 3 TD as stochastic approximation
- 4 Assumptions
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- 6 Where do we converge?
- 7 A worked example
- 8 When it breaks: off-policy
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Setting

Fix a policy π . It induces a **Markov reward process**:

$$P(s' | s) = \sum_a \pi(a | s) P(s' | s, a), \quad r(s) \in [0, 1].$$

Goal: estimate the value function

$$V(s) = \mathbb{E} \left[\sum_{t \geq 0} \gamma^t r(s_t) \mid s_0 = s \right].$$

The Bellman equation

V is the unique fixed point of the **Bellman operator**

$$TV = r + \gamma PV.$$

T is a γ -contraction in $\|\cdot\|_\infty$.

In the tabular case, TD learning is a stochastic version of $V \leftarrow TV$.

Linear function approximation

$|S|$ is huge, so we approximate

$$V_{\theta}(s) = \phi(s)^{\top} \theta, \quad \theta \in \mathbb{R}^d, \quad d \ll |S|.$$

In matrix form, with feature matrix $\Phi \in \mathbb{R}^{|S| \times d}$:

$$V_{\theta} = \Phi \theta.$$

We can only represent functions in $\text{span}(\Phi)$.

Examples of features

Features	What $\text{span}(\Phi)$ contains
One-hot e_s	every V (tabular, exact)
State aggregation	piecewise-constant functions
Tile coding, RBFs	smooth local functions
Last layer of a network	whatever the network learned

Tabular TD is a special case, so every result here covers it.

The update

Observe a transition (s_t, r_t, s_{t+1}) and form the **TD error**:

$$\delta_t = r_t + \gamma \phi(s_{t+1})^\top \theta_t - \phi(s_t)^\top \theta_t.$$

TD(0):

$$\theta_{t+1} = \theta_t + \alpha_t \delta_t \phi(s_t).$$

Compare with Monte Carlo regression

Monte Carlo: regress on the observed return G_t :

$$\theta_{t+1} = \theta_t + \alpha_t (G_t - \phi(s_t)^\top \theta_t) \phi(s_t)$$

This is **SGD** on $\frac{1}{2} \|\Phi\theta - V\|_D^2$.

TD: replaces G_t by the *bootstrapped* target $r_t + \gamma \phi(s_{t+1})^\top \theta_t$.

Lower variance, fully online ... but the target depends on θ .

TD is a *semi*-gradient method

TD differentiates only the prediction $\phi(s_t)^\top \theta$,
not the target $\gamma \phi(s_{t+1})^\top \theta$.

TD is not SGD on any objective: its expected update is $b - A\theta$
with A *non-symmetric* (next section).

So we cannot use the “smoothness + gradient” template from
policy gradient.

Writing TD as a linear system

The update is linear in θ :

$$\theta_{t+1} = \theta_t + \alpha_t (b_t - A_t \theta_t),$$

$$A_t = \phi(s_t)(\phi(s_t) - \gamma\phi(s_{t+1}))^\top, \quad b_t = r_t \phi(s_t).$$

The expected update

With $s_t \sim \mu$ and $D = \text{diag}(\mu)$:

$$A = \mathbb{E}[A_t] = \Phi^\top D(I - \gamma P)\Phi, \quad b = \mathbb{E}[b_t] = \Phi^\top Dr.$$

TD is **linear stochastic approximation**:

$$\theta_{t+1} = \theta_t + \alpha_t (b - A\theta_t + \text{noise}_t).$$

Candidate limit: the solution of $A\theta^* = b$.

The ODE view

With small steps, the iterates track the ODE

$$\dot{\theta} = b - A\theta.$$

This is stable **if all eigenvalues of A have positive real part.**

Sufficient: A is **positive definite**, $\theta^\top A\theta > 0$ for $\theta \neq 0$.

Four assumptions

- 1 **Ergodicity:** the chain P is irreducible and aperiodic, with stationary distribution μ , $\mu(s) > 0$
- 2 **On-policy sampling:** states are drawn from μ (i.i.d., or along the trajectory)
- 3 **Features:** Φ has full column rank, $\|\phi(s)\| \leq 1$
- 4 **Step sizes:** $\sum_t \alpha_t = \infty$, $\sum_t \alpha_t^2 < \infty$
(or small constant steps for finite-time bounds)

What each assumption buys

Assumption	Used for
Ergodicity	μ exists, noise averages out
On-policy	$\mu^P = \mu \Rightarrow A$ positive definite
Full rank	θ^* unique, $\omega = \lambda_{\min}(\Phi^\top D \Phi) > 0$
Bounded features	bounded noise, safe step sizes

On-policy is the one that matters most.

P is non-expansive in the μ -norm

Define $\|V\|_D^2 = \sum_s \mu(s) V(s)^2$.

Lemma. If $\mu P = \mu$, then $\|PV\|_D \leq \|V\|_D$.

Proof. By Jensen, then stationarity:

$$\|PV\|_D^2 = \sum_s \mu(s) \left(\sum_{s'} P(s'|s) V(s') \right)^2 \leq \sum_{s'} \underbrace{\sum_s \mu(s) P(s'|s)}_{=\mu(s')} V(s')^2.$$

A is positive definite

Let $v = \Phi\theta$. Using the lemma and Cauchy–Schwarz:

$$\theta^\top A \theta = \|v\|_D^2 - \gamma \langle v, Pv \rangle_D \geq \|v\|_D^2 - \gamma \|v\|_D \|Pv\|_D \geq (1 - \gamma) \|v\|_D^2.$$

With $\omega = \lambda_{\min}(\Phi^\top D \Phi) > 0$:

$$\theta^\top A \theta \geq (1 - \gamma) \omega \|\theta\|^2$$

Interpretation

$g(\theta) = A\theta - b$ is **strongly monotone**:

$$\langle g(\theta), \theta - \theta^* \rangle \geq (1 - \gamma) \omega \|\theta - \theta^*\|^2.$$

This plays the role of **strong convexity** in SGD, even though g is not the gradient of anything.

Modulus: $(1 - \gamma)\omega$. Small ω or $\gamma \rightarrow 1 \Rightarrow$ slow.

The TD fixed point

Let Π_D be the $\|\cdot\|_D$ -projection onto $\text{span}(\Phi)$:

$$\Pi_D = \Phi(\Phi^\top D \Phi)^{-1} \Phi^\top D.$$

Claim. $A\theta^* = b \iff$

$$\Phi\theta^* = \Pi_D T(\Phi\theta^*)$$

the **projected Bellman equation**.

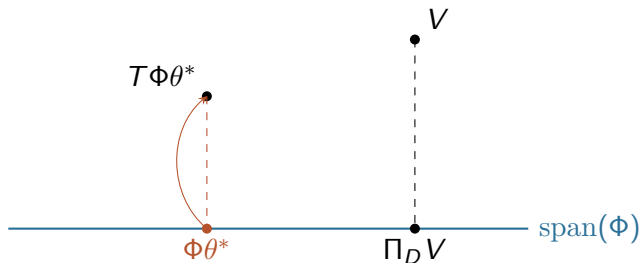
Why: the TD error is orthogonal to the features

$$b - A\theta = \Phi^\top D(r + \gamma P\Phi\theta - \Phi\theta) = \Phi^\top D(T\Phi\theta - \Phi\theta).$$

So $b - A\theta^* = 0$ says the Bellman residual $T\Phi\theta^* - \Phi\theta^*$ is **D-orthogonal** to $\text{span}(\Phi)$.

That is exactly $\Phi\theta^* = \Pi_D T\Phi\theta^*$.

Geometry of the TD fixed point



Apply T , project back: you land where you started.
 $\Phi\theta^*$ is **not** the best approximation $\Pi_D V$ in general.

$\Pi_D T$ is a contraction

Projection is non-expansive, and by the key lemma:

$$\begin{aligned}\|\Pi_D TV - \Pi_D TV'\|_D &\leq \|TV - TV'\|_D = \gamma \|P(V - V')\|_D \\ &\leq \gamma \|V - V'\|_D.\end{aligned}$$

So the TD fixed point **exists and is unique**.

Note: this uses $\|\cdot\|_D$ with the *on-policy* μ .

How good is the TD fixed point?

Theorem (Tsitsiklis & Van Roy, 1997).

$$\|\Phi\theta^* - V\|_D \leq \frac{1}{\sqrt{1-\gamma^2}} \|\Pi_D V - V\|_D.$$

- Tabular features: $\Pi_D V = V$, so TD is exact
- Otherwise: at most a factor $\frac{1}{\sqrt{1-\gamma^2}}$ worse than the best fit
- TD(λ): factor $\frac{1-\lambda\gamma}{1-\gamma}$, which is 1 at $\lambda = 1$ (Monte Carlo)

Proof: Pythagoras

$\Phi\theta^* - \Pi_D V \in \text{span}(\Phi)$ and $\Pi_D V - V \perp_D \text{span}(\Phi)$:

$$\|\Phi\theta^* - V\|_D^2 = \|\Phi\theta^* - \Pi_D V\|_D^2 + \|\Pi_D V - V\|_D^2.$$

Using $\Phi\theta^* = \Pi_D T\Phi\theta^*$ and $\Pi_D V = \Pi_D TV$:

$$\|\Phi\theta^* - \Pi_D V\|_D \leq \gamma \|\Phi\theta^* - V\|_D.$$

Rearrange: $(1 - \gamma^2) \|\Phi\theta^* - V\|_D^2 \leq \|\Pi_D V - V\|_D^2.$

□

Two states, one feature

States s_1, s_2 ; from either state, move to s_1 or s_2 w.p. $\frac{1}{2}$.

$$r = (1, 0), \quad \gamma = 0.9, \quad \mu = \left(\frac{1}{2}, \frac{1}{2}\right).$$

True values: $V = (5.5, 4.5)$.

Feature: $\phi(s_1) = 1, \phi(s_2) = 2$, so $V_\theta = (\theta, 2\theta)$.

Computing the TD fixed point

$$\Phi^\top D\Phi = \frac{1}{2}(1 + 4) = 2.5, \quad \Phi^\top DP\Phi = \frac{1}{2}(1.5 + 3) = 2.25$$

$$A = 2.5 - 0.9 \cdot 2.25 = 0.475, \quad b = \frac{1}{2} \cdot 1 = 0.5$$

$$\theta^* = \frac{b}{A} = \frac{20}{19} \approx 1.05$$

$A > 0$, as the key lemma promised.

TD fixed point vs. best approximation

	θ	V_θ	$\ V_\theta - V\ _D$
Best fit $\Pi_D V$	2.90	(2.90, 5.80)	2.06
TD fixed point	1.05	(1.05, 2.11)	3.57
Bound: $\frac{2.06}{\sqrt{1-0.81}}$			4.72

Poor features force $V(s_2) = 2V(s_1)$; bootstrapping makes it worse.

The $\theta \rightarrow 2\theta$ example

Same features $\phi(s_1) = 1$, $\phi(s_2) = 2$.

Only update on the transition $s_1 \rightarrow s_2$ with reward 0:

$$\theta \leftarrow \theta + \alpha (\gamma \cdot 2\theta - \theta) \cdot 1 = (1 + \alpha(2\gamma - 1)) \theta.$$

For $\gamma > \frac{1}{2}$: $|\theta_t| \rightarrow \infty$ for every $\alpha > 0$.

The sampling distribution is not stationary for $P \Rightarrow A$ is not positive definite.

The deadly triad

Divergence can occur when all three are present:

- ① Function approximation
- ② Bootstrapping
- ③ Off-policy sampling

Fixes: gradient-TD (GTD2, TDC), emphatic TD, target networks and regularization in practice.

Asymptotic convergence

Theorem (Tsitsiklis & Van Roy, 1997).

Under the four assumptions with Robbins–Monro steps,

$$\theta_t \rightarrow \theta^* \quad \text{almost surely,}$$

even with Markov (trajectory) sampling.

Proof: ODE method + positive definiteness of A .

Finite-time analysis: setup

Assume i.i.d. sampling $s_t \sim \mu$ for now. Let $e_t = \theta_t - \theta^*$.

Write the update direction as

$$g_t(\theta) = A_t \theta - b_t = A_t(\theta - \theta^*) + \underbrace{(A_t \theta^* - b_t)}_{\text{noise at } \theta^*}$$

$$\sigma^2 = \mathbb{E} \|A_t \theta^* - b_t\|^2, \quad \text{and } \|A_t\| \leq 1 + \gamma \leq 2.$$

One step

$$\|e_{t+1}\|^2 = \|e_t\|^2 - 2\alpha \langle g_t, e_t \rangle + \alpha^2 \|g_t\|^2$$

In expectation, using the key lemma and $\|A_t\| \leq 2$:

$$\mathbb{E}\|e_{t+1}\|^2 \leq (1 - 2\alpha(1 - \gamma)\omega + 8\alpha^2) \mathbb{E}\|e_t\|^2 + 2\alpha^2\sigma^2$$

Same recursion as SGD on a strongly convex function.

Constant step size

With $\alpha \leq (1 - \gamma)\omega/8$:

$$\mathbb{E}\|e_{t+1}\|^2 \leq (1 - \alpha(1 - \gamma)\omega) \mathbb{E}\|e_t\|^2 + 2\alpha^2\sigma^2.$$

Unroll:

$$\mathbb{E}\|\theta_T - \theta^*\|^2 \leq \underbrace{(1 - \alpha(1 - \gamma)\omega)^T \|\theta_0 - \theta^*\|^2}_{\text{bias: linear decay}} + \underbrace{\frac{2\alpha\sigma^2}{(1 - \gamma)\omega}}_{\text{noise floor}}$$

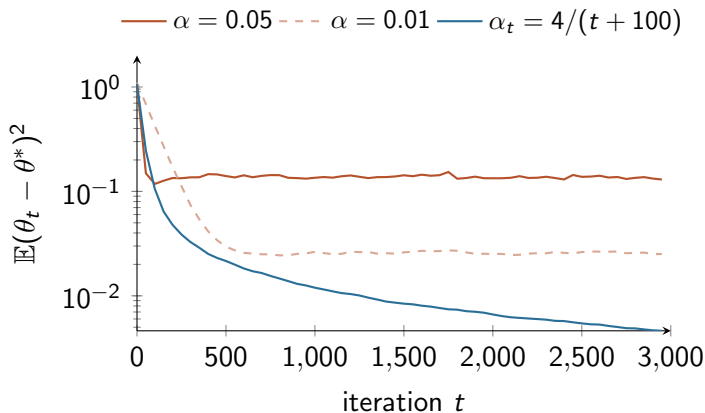
Decaying step size

With $\alpha_t \asymp \frac{1}{(1-\gamma)\omega t}$:

$$\mathbb{E}\|\theta_T - \theta^*\|^2 = O\left(\frac{\sigma^2}{(1-\gamma)^2\omega^2 T}\right).$$

- The constant in α_t must be large enough (as in SGD)
- Averaging the iterates (Polyak–Ruppert) removes this sensitivity

Simulation on the two-state example



Larger constant step: faster, but a higher floor. Decaying step: keeps improving.

Markov (trajectory) sampling

Consecutive states are correlated, so $\mathbb{E}[g_t \mid \theta_t] \neq g(\theta_t)$.

Fix: compare with the chain τ steps back, where $\tau \approx$ **mixing time**.

Constant step (schematic):

$$\mathbb{E}\|\theta_T - \theta^*\|^2 \lesssim (1 - c\alpha)^{T-\tau} + \alpha\tau \cdot \frac{\sigma^2}{(1-\gamma)\omega}$$

Same shape as i.i.d., with the noise floor inflated by τ .

Rates at a glance

Setting	Step size	Guarantee
Any	Robbins–Monro	$\theta_t \rightarrow \theta^*$ a.s.
i.i.d.	constant α	linear, to floor $O(\alpha)$
i.i.d.	$\asymp 1/t$	$O(1/T)$
Markov	constant α	linear, to floor $O(\alpha \tau)$
Markov	$\asymp 1/t$	$\tilde{O}(\tau/T)$

All constants scale with the modulus $(1 - \gamma)\omega$.

Three algorithms, one proof template

	Structure	Rate
SGD, strongly convex	strong convexity	$O(1/T)$
Policy gradient	nonconvex + gradient domination	$O(1/t)$ (exact)
TD, linear	strongly monotone, not a gradient	$O(1/T)$

What matters is $\langle g(\theta), \theta - \theta^* \rangle \geq c \|\theta - \theta^*\|^2$,
not whether g is a gradient.

Takeaways

- ① TD is linear stochastic approximation, not SGD
- ② On-policy sampling $\Rightarrow A$ positive definite $\Rightarrow$ strongly monotone
- ③ TD converges to the **projected Bellman fixed point**, within $\frac{1}{\sqrt{1-\gamma^2}}$ of the best approximation
- ④ Rates match strongly convex SGD, with modulus $(1 - \gamma)\omega$
- ⑤ Off-policy breaks positive definiteness $\Rightarrow$ possible divergence

References I

- Sutton (1988). Learning to predict by the methods of temporal differences.
- Baird (1995). Residual algorithms: reinforcement learning with function approximation.
- Tsitsiklis & Van Roy (1997). An analysis of temporal-difference learning with function approximation.
- Borkar & Meyn (2000). The ODE method for convergence of stochastic approximation and reinforcement learning.
- Sutton, Maei, et al. (2009). Fast gradient-descent methods for temporal-difference learning with linear function approximation.
- Dalal, Szörényi, Thoppe, Mannor (2018). Finite sample analyses for TD(0) with function approximation.
- Lakshminarayanan & Szepesvári (2018). Linear stochastic approximation: how far does constant step-size and iterate averaging go?
- Bhandari, Russo, Singal (2018). A finite time analysis of temporal difference learning with linear function approximation.

References II

- Srikant & Ying (2019). Finite-time error bounds for linear stochastic approximation and TD learning.

Overview

- 1 TD-Learning with Linear Function Approximation
- 2 Policy Gradient

Roadmap

- 1 The problem
- 2 The policy gradient
- 3 Softmax and Gaussian policies
- 4 Policy gradient *is* SGD
- 5 Assumptions
- 6 Stationary points
- 7 Global optimality
- 8 Rates and takeaways

Setting

Discounted MDP $(\mathcal{S}, \mathcal{A}, P, r, \gamma, \rho)$ with $r \in [0, 1]$.

A **stochastic policy** $\pi_{\theta}(a \mid s)$ with parameters $\theta \in \mathbb{R}^d$.

Objective:

$$J(\theta) = \mathbb{E}_{\tau \sim \pi_{\theta}} [R(\tau)], \quad R(\tau) = \sum_{t \geq 0} \gamma^t r(s_t, a_t).$$

Why is this hard?

$$\nabla_{\theta} \mathbb{E}_{\tau \sim \pi_{\theta}} [R(\tau)]$$

θ changes the **distribution** we sample from,
not the function inside the expectation.

We cannot just differentiate inside the expectation.

The log-derivative trick

Use $\nabla p_\theta = p_\theta \nabla \log p_\theta$:

$$\nabla J(\theta) = \int R(\tau) \nabla p_\theta(\tau) d\tau = \mathbb{E}_{\tau \sim \pi_\theta} [R(\tau) \nabla \log p_\theta(\tau)].$$

An expectation under $\pi_\theta \Rightarrow$ we can **sample** it.

The dynamics drop out

$$\log p_{\theta}(\tau) = \log \rho(s_0) + \sum_t \log \pi_{\theta}(a_t \mid s_t) + \sum_t \log P(s_{t+1} \mid s_t, a_t)$$

Only the middle term depends on θ , so

$$\nabla J(\theta) = \mathbb{E}_{\tau} \left[R(\tau) \sum_t \nabla_{\theta} \log \pi_{\theta}(a_t \mid s_t) \right].$$

Model-free: no knowledge of P needed.

Policy gradient theorem

$$\nabla J(\theta) = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d^\pi, a \sim \pi} [\nabla_\theta \log \pi_\theta(a \mid s) A^\pi(s, a)]$$

- d^π : discounted state-visitation distribution
- $A^\pi = Q^\pi - V^\pi$: advantage

Intuition: raise the probability of actions that did better than average.

Baselines

For any $b(s)$:

$$\mathbb{E}_{a \sim \pi} [\nabla_{\theta} \log \pi_{\theta}(a | s) b(s)] = 0.$$

So subtracting a baseline keeps the gradient **unbiased** but can greatly reduce its **variance**.

Same idea as *control variates*.

The parameterization *is* the approximation

	Policy class	Contains π^* ?
Tabular	one parameter per (s, a)	yes (in the limit)
Linear	$\theta^\top \phi(s, a)$	only if features allow
Neural	$\text{NN}_\theta(s, a)$	approximately

A restricted class shows up later as an **approximation error** in the rates.

Example 1: Softmax policy

For discrete actions:

$$\pi_{\theta}(a \mid s) = \frac{\exp f_{\theta}(s, a)}{\sum_b \exp f_{\theta}(s, b)}$$

Score function:

$$\nabla_{\theta} \log \pi_{\theta}(a \mid s) = \nabla f_{\theta}(s, a) - \mathbb{E}_{b \sim \pi_{\theta}} [\nabla f_{\theta}(s, b)]$$

“Features of the chosen action minus the average features.”

Softmax: special cases

Linear: $f_\theta = \theta^\top \phi(s, a)$

$$\nabla_\theta \log \pi_\theta(a | s) = \phi(s, a) - \bar{\phi}_\pi(s)$$

Tabular: $f_\theta = \theta_{s,a}$

$$\nabla_\theta \log \pi_\theta(a | s) = e_a - \pi_\theta(\cdot | s) \quad (\text{in row } s)$$

Bounded: $\|\nabla_\theta \log \pi_\theta\| \leq 2 \max \|\phi\|$.

Worked example: two-armed bandit

One state, actions $\{1, 2\}$, rewards $r_1 = 1$, $r_2 = 0$.

$$J(\theta) = \pi_1 = \frac{e^{\theta_1}}{e^{\theta_1} + e^{\theta_2}}$$

Exact gradient: $\partial J / \partial \theta_a = \pi_a(r_a - J)$, so

$$\nabla J = (\pi_1(1 - \pi_1), -\pi_1(1 - \pi_1)).$$

At $\theta = (0, 0)$: $\nabla J = (\frac{1}{4}, -\frac{1}{4})$.

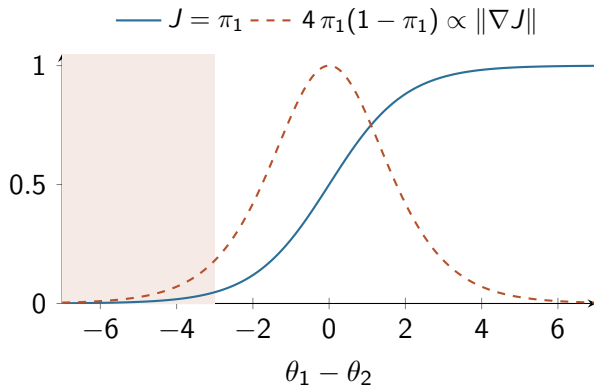
Two-armed bandit: the REINFORCE estimate

Sample one action, use $\hat{g} = r_a (e_a - \pi)$. At $\pi = (\frac{1}{2}, \frac{1}{2})$:

- $a = 1$ (prob. $\frac{1}{2}$): $\hat{g} = (\frac{1}{2}, -\frac{1}{2})$
- $a = 2$ (prob. $\frac{1}{2}$): $\hat{g} = (0, 0)$

$$\mathbb{E}[\hat{g}] = \frac{1}{2}(\frac{1}{2}, -\frac{1}{2}) = (\frac{1}{4}, -\frac{1}{4}) = \nabla J \quad \checkmark$$

Two-armed bandit: flat regions



Bad initialization ($\pi_1 \approx 0$): far from optimal, yet gradient ≈ 0 .

Example 2: Gaussian policy

For continuous actions:

$$\pi_{\theta}(a | s) = \mathcal{N}(\mu_{\theta}(s), \sigma^2), \quad \mu_{\theta}(s) = \theta^{\top} \phi(s)$$

Score function:

$$\nabla_{\theta} \log \pi_{\theta}(a | s) = \frac{a - \mu_{\theta}(s)}{\sigma^2} \phi(s)$$

“Action above the mean and good reward $\Rightarrow$ move the mean up.”

Worked example: quadratic bandit

$$a \sim \mathcal{N}(\theta, \sigma^2), \quad r(a) = -(a - a^*)^2, \quad \Delta = \theta - a^*.$$

$$J(\theta) = -\Delta^2 - \sigma^2, \quad \nabla J = -2\Delta.$$

Write $a = \theta + \sigma z$, $z \sim \mathcal{N}(0, 1)$:

$$\hat{g} = \frac{z}{\sigma} r(a), \quad \mathbb{E}[\hat{g}] = -2\Delta \quad \checkmark$$

Quadratic bandit: variance

Without baseline:

$$\text{Var}(\hat{g}) = \frac{\Delta^4}{\sigma^2} + 14\Delta^2 + 15\sigma^2$$

With baseline $b = J(\theta)$:

$$\text{Var}(\hat{g}_b) = 8\Delta^2 + 10\sigma^2$$

- Less exploration ($\sigma \rightarrow 0$) $\Rightarrow$ noisier gradients
- The baseline removes the blow-up term

Gaussian: what about the assumptions?

The score is **not** uniformly bounded: a is unbounded.

Only its second moment is bounded:

$$\mathbb{E} \|\nabla_{\theta} \log \pi_{\theta}\|^2 = \frac{\|\phi\|^2}{\sigma^2}$$

So constants scale like $G \sim 1/\sigma$ and $L \sim 1/\sigma^2$:
small noise $\Rightarrow$ worse smoothness $\Rightarrow$ smaller step sizes.

Two examples side by side

	Softmax	Gaussian
Score	$\phi(s, a) - \bar{\phi}_\pi(s)$	$\frac{a - \mu}{\sigma^2} \phi(s)$
Bounded score	yes	in expectation, $\sim 1/\sigma$
Failure mode	vanishing gradient	high variance

Both failure modes return when we discuss rates.

The update

Sample a few trajectories $\Rightarrow$ unbiased $\hat{g}_t$ with $\mathbb{E}[\hat{g}_t] = \nabla J(\theta_t)$.

$$\theta_{t+1} = \theta_t + \eta \hat{g}_t$$

This is **stochastic gradient ascent**.

A dictionary

Supervised learning	Policy gradient
$F(\theta) = \mathbb{E}_z[f(\theta; z)]$	$J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta}[R(\tau)]$
minibatch gradient	rollouts + score function
minibatches, SVRG	baselines, critics, VR-PG
Adam, natural gradient	NPG, TRPO, PPO

The one real difference

SGD: data distribution is **fixed**.

PG: data distribution **moves with θ** .

In the analysis, this is absorbed into the **smoothness constant**.

Three assumptions

- 1 **Bounded rewards:** $r \in [0, 1]$, so returns $\leq \frac{1}{1-\gamma}$
- 2 **Regular policy:** $\|\nabla_{\theta} \log \pi_{\theta}\| \leq G$, $\|\nabla_{\theta}^2 \log \pi_{\theta}\| \leq M$
- 3 **Bounded variance:** $\mathbb{E}\|\hat{g} - \nabla J\|^2 \leq \sigma^2$

Truncating rollouts at horizon H adds only $O(\gamma^H)$ bias.

Key lemma: smoothness

Under assumptions 1–2, J is **L -smooth**:

$$\|\nabla J(\theta) - \nabla J(\theta')\| \leq L\|\theta - \theta'\|,$$

$$L = O\left(\frac{M}{(1-\gamma)^2} + \frac{G^2}{(1-\gamma)^3}\right).$$

Tabular softmax: $L = 8/(1-\gamma)^3$.

Step 1: the descent lemma

By L -smoothness, for the ascent step $\theta_{t+1} = \theta_t + \eta \hat{g}_t$:

$$J(\theta_{t+1}) \geq J(\theta_t) + \eta \langle \nabla J, \hat{g}_t \rangle - \frac{L\eta^2}{2} \|\hat{g}_t\|^2.$$

Take expectations, use $\mathbb{E}\hat{g}_t = \nabla J$ and $\eta \leq 1/L$:

$$\mathbb{E}J(\theta_{t+1}) \geq \mathbb{E}J(\theta_t) + \frac{\eta}{2} \mathbb{E}\|\nabla J(\theta_t)\|^2 - \frac{L\eta^2\sigma^2}{2}.$$

Step 2: telescope

Sum over $t = 0, \dots, T - 1$ and use $J \leq J^*$:

$$\frac{1}{T} \sum_{t < T} \mathbb{E} \|\nabla J(\theta_t)\|^2 \leq \frac{2(J^* - J(\theta_0))}{\eta T} + L\eta\sigma^2.$$

With $\eta \sim 1/\sqrt{T}$:

$$\min_{t < T} \mathbb{E} \|\nabla J(\theta_t)\|^2 = O\left(\frac{1}{\sqrt{T}}\right).$$

Sample complexity

- $\mathbb{E}\|\nabla J\| \leq \epsilon$ needs $O(\epsilon^{-4})$ samples
- Variance reduction (SVRPG, STORM-type): $O(\epsilon^{-3})$

Exactly the nonconvex SGD rates.

But: this only guarantees a *stationary point*.

J is nonconvex

$J(\theta)$ is non-concave, even for the tabular parameterization.

For general nonconvex functions: stationary $\nRightarrow$ optimal.

RL has extra structure that saves us.

Performance difference lemma

For any two policies π, π' :

$$J(\pi') - J(\pi) = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d^{\pi'}} \mathbb{E}_{a \sim \pi'} [A^{\pi}(s, a)].$$

Set $\pi' = \pi^*$: if π is suboptimal, some action has **positive advantage** somewhere, and the gradient sees it.

Gradient domination

This yields a Polyak–Łojasiewicz-type inequality (schematic):

$$J^* - J(\theta) \leq C \cdot \|\nabla J(\theta)\|$$

Every stationary point is globally optimal.

Same mechanism as global convergence of GD on PL functions (least squares, some over-parameterized networks).

From domination to a rate

Exact gradients, $\eta = 1/L$, gap $\delta_t = J^* - J(\theta_t)$:

$$\delta_{t+1} \leq \delta_t - \frac{1}{2L} \|\nabla J(\theta_t)\|^2 \leq \delta_t - \frac{\delta_t^2}{2LC^2}$$

$$\Rightarrow \delta_t = O\left(\frac{LC^2}{t}\right)$$

Where exploration enters

The constant C contains the **distribution mismatch coefficient**:

$$\left\| \frac{d^{\pi^*}}{\mu} \right\|_{\infty}$$

The start distribution μ must cover the states π^* visits.

Exploration is not free: it appears directly in the constant.

Rates: exact gradients

Method	Rate	Reason
Softmax PG	$O(1/t)$	smooth + domination
Softmax PG, worst case	exp. in $ \mathcal{S} $	vanishing gradients
Entropy-regularized PG	linear	strong PL constant
Natural PG	$O(1/t)$, dim-free	KL geometry
NPG, growing steps	linear	→ policy iteration

Rates: stochastic gradients

Goal	Samples
ϵ -stationary, vanilla PG	$O(\epsilon^{-4})$
ϵ -stationary, variance-reduced	$O(\epsilon^{-3})$
ϵ -optimal, Fisher non-degenerate	$\tilde{O}(\epsilon^{-3})$ to $\tilde{O}(\epsilon^{-2})$

Global results carry an **approximation-error floor** from the policy class.

Why softmax PG can be slow

Recall the bandit: $\|\nabla J\| \propto \pi(a^*)(1 - \pi(a^*))$.

Euclidean geometry in θ interacts badly with the simplex.

Fixes:

- Entropy regularization: keeps $\pi(a^*)$ away from 0
- Natural gradient: KL geometry, update

$$\pi_{t+1} \propto \pi_t \exp\left(\frac{\eta}{1-\gamma} A^{\pi_t}\right)$$

Takeaways

- ➊ Policy gradient = SGD with a score-function estimator
- ➋ Smoothness + bounded variance $\Rightarrow$ SGD rates for stationarity
- ➌ Performance difference lemma $\Rightarrow$ gradient domination
 $\Rightarrow$ stationary points are global
- ➍ Hard parts: **exploration** (mismatch) and **geometry** (flat regions)
- ➎ Regularization and natural gradients $\Rightarrow$ TRPO, PPO

References I

- Williams (1992). Simple statistical gradient-following algorithms for connectionist RL.
- Sutton, McAllester, Singh, Mansour (2000). Policy gradient methods for RL with function approximation.
- Kakade (2002). A natural policy gradient.
- Papini et al. (2018). Stochastic variance-reduced policy gradient.
- Agarwal, Kakade, Lee, Mahajan (2021). On the theory of policy gradient methods.
- Mei, Xiao, Szepesvári, Schuurmans (2020). On the global convergence rates of softmax policy gradient methods.
- Li, Wei, Chi, Gu, Chen (2021). Softmax policy gradient methods can take exponential time to converge.
- Cen, Cheng, Chen, Wei, Chi (2021). Fast global convergence of natural policy gradient methods with entropy regularization.
- Lan (2021). Policy mirror descent for RL.

References II

- Xiao (2022). On the convergence rates of policy gradient methods.
- Yuan, Gower, Lazaric, Xiao (2022). A general sample complexity analysis of vanilla policy gradient.
- Fatkhullin, Barakat, Kireeva, He (2023). Stochastic policy gradient methods: improved sample complexity for Fisher-non-degenerate policies.